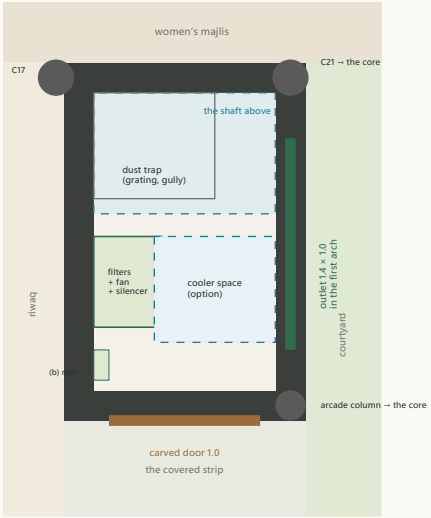
Air system study v4 · the malqaf at L5: a large courtyard breeze and a filtered supply · Taif / Palm House V10.0

The owner's decisions of 9 Nov 2026 (D1–D11, `studies/malqaf_L5.typ`, adopted): the breeze at the seating is the goal · no HRV: a filtered supply fan · one tower, two paths after the head · no water tank on the tower · a dust-tight house, the courtyard an open outdoor room · the tower touches no bedroom · render and paint, no stone · simple controls. **Every size and performance figure is ASSUMED, to be designed by the MEP engineer** compiling this study is not engineering approval.



the GF foot chamber · plan · 1 m = 2.0 cm

Air system study v4 · checks passed (9)

1 · The concept

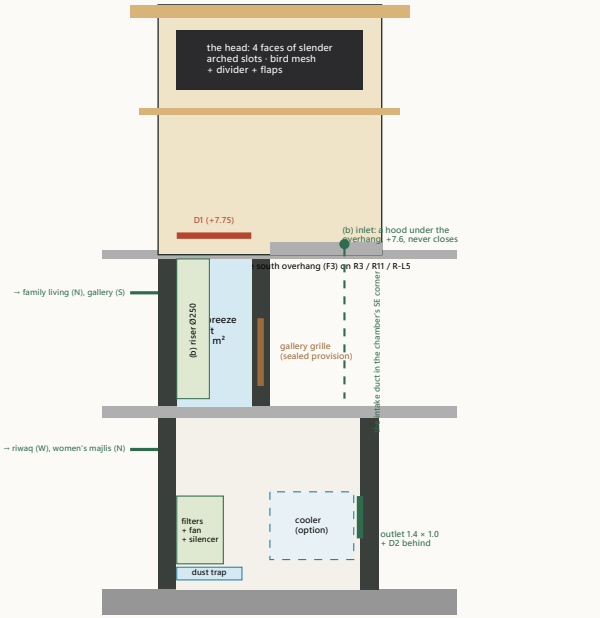
- **One tower at the courtyard's north-west corner** (L5: `studies/malqaf_L5.typ`, the owner's decision after a location study): an **independent RC core** from its own pad to the head, inside the covered strip under the first-floor projection, against the women's majlis's wall (nothing taken from the majlis or the family living); **no bedroom touches it** (D10). It catches Taif's warm-season evening wind (W ≈ 50 %, NW ≈ 30 %, ≈ 21 km/h; the climate basis in `studies/malqaf_L5.typ` Phase 0) above the roof's own wake.
- **Path (a), the courtyard breeze (D1, the goal):** the head's four faces → the + divider and the back-draught flaps → **D1** → the shaft (0.84 m²) → the **dust trap** at its foot → the chamber → the **outlet 1.4 × 1.0** raised to +0.60…+1.60 with its louvres fixed 15° south-east and **D2**, in the **first arch of the arcade**: over the water trough, under the olive, then the seating (H2, 10 Nov 2026): **no channel**. Unfiltered by design (a bird mesh only).
- **Path (b), the house's fresh air (no HRV, D2):** a **never-closing** downward inlet under the head's south overhang → a washable pre-filter + a fine filter → a quiet **EC fan** (750 m³/h) → a silencer → a sealed riser → high grilles into the **family living**, the **gallery**, the **riwaq** and the **women's majlis**; the house runs at a slight **positive pressure**, dust stays out (D5). **The kitchen hood keeps its own interlocked make-up inlet (mandatory).**
- **Simple controls (owner):** two wall switches, one dust-sensor relay, spring-return dampers (§ 5). **The cooler (D9):** an option: its space, a power point and a floor drain in the chamber; the water point capped; the malqaf is a **dry system**.

2 · The core: the shaft, the foot chamber, the door

- **The shaft** (FF and above): outer 1.6 × 1.2 m, **0.96 m² clear**; the sealed Ø250 riser of path (b) in its west side, the breeze part **0.84 m²**; smooth-rendered inside; the gallery's turn past bedroom 2's bath stays **1.5 m clear**. On its south face at the FF a carved arched timber grille 0.8 × 1.6 (sill +4.2): **a sealed provision** (owner), no damper now.
- **The foot chamber** (GF): outer 1.6 × 2.37 m to the first arcade column, **2.36 m² clear**, ≈ 3.1 m high, open above into the shaft: the dust trap (a 0.8 × 0.7 GRP basin 0.3 m deep under the shaft, a lift-out grating, a trapped gully with a trap primer); the filter cabinet, the fan and the silencer on the west wall; the cooler's space at the outlet; a floor fall to the gully, the walls tanked to +0.30, IP44 sockets on their own circuit, a capped water point. **The door:** a carved timber door **1 × 2.4** with a semicircular head, on the south face onto the covered strip: the cooler rolls in, every cleaning task passes here.
- **Absorbed in the core:** the columns C17 (4.25, 6.5) and C21 (5.8, 6.5) and the first arcade column (5.8, 8.667): the core is the **fourth support** of the west projection (B3 / B4 first spans 6.50 → 4.33 m; `studies/structure.typ`).

3 · The two paths in section · the head · the dampers · the intake and the exhausts

SECTION N–S through the core (y 5.8–9.6, stretched × 2.2) · levels from the GF floor



3.1 · The head (F3; render and paint, no stone)

- Outer **1.6 × 2.4 m** on the shaft and the gallery roof, an **RC frame** (0.30 corner piers, ring beams at the band and the crown), the slender ribs in rendered block; a light sand render, the band (+10.6), the two-step Hijazi crown (+12.9 … +13.2) and the slot surrounds in the calm honey tone, every projection with a drip. **Openings +11.2 … +12.6** on all four faces: 3 slots on the short faces, 5 on the long (W / E), each 0.2 m wide with a small semicircular top; free ≈ 0.83 / 1.38 m² per face. The W and N faces are the catchers (Phase 0); the openings start above the roof's recirculation bubble. Inside: a 20 mm bird mesh in frames lowered on cords to a 0.5 m hatch in the west face (+7.6 … +8.1), the + divider and four back-draught flaps (the windward quarter feeds, the leeward shut). **Crown +13.2**: 4.5 m above the attic coping; seen whole from the courtyard seating, 1.9 m from the street at 14 m, **not** from the gate (accepted by the owner).

3.2 · The dampers (two; insulated, low-leakage, spring-return)

- **D1** at the shaft's head (+7.75, 1.2 × 0.8) and **D2** behind the outlet's louvres: both close on power loss; driven together by the one switch MALQAF and closed by the dust relay. **No D3:** the gallery grille is a sealed provision. **D4:** a passive gravity louvre Ø160 from the chamber to the outlet (no motor), so the chamber never stays damp. The hood's make-up damper: its own, interlocked with the hood (**mandatory**).

3.3 · The intake and the exhausts (ASHRAE 62.1-2022 Table 5-1 allowances)

Pair	Distance	Need	Basis
breeze head ↔ kitchen hood outlet	12.6 m	≥ 4.6	class 3 (cooking) → ≥ 4.6 m; downwind, 9 m below
breeze head ↔ east stack's vent	8.7 m	≥ 3	vents → ≥ 3 m; downwind
breeze head ↔ west stack's vent (re-routed SE)	10.1 m	≥ 3	vents → ≥ 3 m; no longer upwind
breeze head ↔ AC condensers (Z2b)	5.2 m	≥ 2	their hot plume rises leeward in a W wind; 3.7 m in plan, 2.6 m below the openings
filtered inlet ↔ hood outlet	10.5 m	≥ 4.6	class 3
filtered inlet ↔ the two vents	8.6 m	≥ 3	≥ 3 m
filtered inlet ↔ LPG cabinet / generator point (the yard)	11.9 m	≥ 4.6	class 3 / 4 allowances; the generator point ≥ 9 m

- The west stack's vent **re-routed in the attic to the south-east strip** (owner, 9 Nov 2026): no exhaust stands upwind of the head in the summer-evening west wind. The AC condensers stay in Z2b, east of the head and below its openings.

4 · The breeze, estimated honestly · 5 · The house ventilation (the filtered supply)

4 · What the courtyard will feel

The path's resistance, referred to a 0.10 m² reference section (**assumed loss coefficients, ASHRAE Fundamentals ch. 21 orders of magnitude**); the V9.3 chain (L1 at the north-east corner, 10.6 m of channel) scored 1.5: the L5 path is **≈ 4.1 ×** its flow in the same wind.

Element	ζ (0.10 m²)
the head: the windward long face, 5 slots + bird mesh (1.24 m² free)	0.02
the back-draught flaps and D1 (0.84 m²)	0.01
the shaft, 3.8 m, smooth render	0
the turn into the chamber over the dust trap	0.02
the outlet louvres (0.84 m² free) and D2	0.04
total	**0.09**

Driving pressure: wind at the head ½ρv²·ΔCp with ρ 1.15 kg/m³ (Taif, 1 700 m) and ΔCp 0.8 (windward face against the courtyard well, assumed), the head at ≈ +12 (≈ 4 % more than the 10 m mast); in still air the stack of the 11 m path with 3 K: 1.24 Pa.

Wind at 10 m	Δp (Pa)	flow (m³/s)	at the louvres (m/s)
calm (stack only)	1.24	0.49	0.6
gentle, 10 km/h	3.84	0.85	1
the summer-evening mean, 21 km/h	16.93	**1.79**	2.1
a strong evening, 30 km/h	34.55	2.56	3.1

- **Honest conclusion:** at the typical summer evening (21 km/h at the mast, Phase 0: ≈ 50 % W, 30 % NW) the outlet delivers ≈ 1.79 m³/s, ≈ 2.1 m/s at the louvres and ≈ 0.5–0.8 m/s at the seating 2 m away: **a clearly felt breeze**, ≈ 4.1 × the L1 system; a gentle 10 km/h wind still gives 0.85 m³/s; a calm evening gives little (the stack 0.49 m³/s). About 1 800 pleasant, dust-free seating hours a year (`studies/malqaf_L5.typ` Phase 0). The louvres are fixed 15° to the south-east (H2; 11 Nov 2026: the olive stands in the jet's path and the trough in front of the sink window, so the jet runs outlet → olive → water): the jet passes under the olive's crown and over the trough, meets the kitchen wall and settles on the seating from the side, ≈ 0.6 m/s on the whole body; the family closes the switch when it is too much.
- **If a cooler is added later (D9, option):** a 90–150 L portable evaporative cooler in the chamber moves ≈ 0.8–1.4 m³/s and cools the dry afternoon air from ≈ 35 to ≈ 20–22 °C (RH 12–20 %); its water ≈ 15–25 L/h; it cools **the people in the jet**, not the open courtyard. Provided now: the space, a power point, the floor drain; the water point capped.

5 · The house's ventilation (no HRV: a filtered supply fan, D2)

- **Preliminary airflow** (ASHRAE 62.2-2022 § 4.1.1: 0.15 L/s·m² of floor + 3.5 L/s per bedroom + 1): conditioned floor ≈ 462 m² clear (502 m² on axes), 5 bedrooms → **325 m³/h normal**; the men's majlis full (24 guests, ASHRAE 62.1 assembly) → **704 m³/h boost**. **The MEP engineer sizes it with the final rooms.**
- **The fan:** an EC plug / centrifugal fan **750 m³/h at 250 Pa** external, ≈ 160 W, in its cabinet in the foot chamber on anti-vibration mounts, a **0.9 m lined silencer** after it (≈ −15 dB), ≈ 38 dB(A) in the chamber: the courtyard hears nothing through the 200 RC wall and the lined cabinet. Two speeds from the switch (normal / boost), set once at commissioning.
- **The filters:** ISO Coarse 60 % (G4) washable 500 × 500 × 48 + ISO ePM1 ≥ 50 % (F7 class) 500 × 500 × 292 a mechanical gauge on the cabinet: change the fine filter at 200 Pa (≈ 6 months); the pre-filter washed monthly. The fine filter holds ≥ 50 % of PM1 (≈ 80 % of PM2.5): during a 150 µg/m³ dust event the house stays at roughly a quarter of the outdoor level (the house tight, D5).
- **Positive pressure:** the fan supplies 325 m³/h into a tight envelope; the extracts (the bathrooms ≈ 150–250 m³/h continuous, the dryer) are smaller: the house runs at ≈ **+5–10 Pa**; every door undercut 15 mm or with a transfer grille. **The kitchen hood** (600–1 000 m³/h) exceeds the fan: without its make-up inlet the house would go negative and pull dust through every crack and the malqaf's grille: **the interlocked make-up inlet (≈ 300 m³/h, filtered, motorised, opens with the hood) is mandatory** (owner, 9 Nov 2026).
- **The supply route:** the inlet hood under the overhang (+7.6) → a Ø315 duct down the chamber's south-east corner → the filters and the fan → the silencer → a sealed Ø250 riser in the shaft's west side → high grilles (+6.4) into the **family living** (through the core's north wall) and the **gallery** (south face), and at the GF (+2.7) into the **riwaq** (west face) and the **women's majlis** (north wall); the other rooms through the riwaq's open plan and the door undercuts. **No bedroom is ducted from the core.**
- **Winter:** the breeze closed (switch 1 off), the fan at normal: ≈ 1.11 kW of heat at a 10 K difference (no heat recovery): small; the family sets boost when the house is full.

6 · Controls (simple, the owner's rule) · 7 · Maintenance · 8 · Decisions and what the engineers check

6 · Controls: two switches, one relay

Switch	ON	OFF
1 · MALQAF	D1 + D2 open: the courtyard breeze, whenever the family finds the evening pleasant	D1 + D2 shut: winter, dust, away. **The dust relay forces OFF** while the air is dirty
2 · HOUSE FAN	boost (704 m^3/h): guests, after a dust day, a full house	normal (325 m^3/h): always on; the filtered path never stops

- **The dust relay:** an outdoor PM2.5 sensor in a weather hood on the head's north face: above **50 µg/m^3 for 10 min** D1 and D2 close by themselves and the switch's lamp shows "dust"; they reopen when the air has been below 25 µg/m^3 for 30 min and the switch is on. **Every damper spring-return:** closed on power loss. **No controller, no modes** (owner, 9 Nov 2026). A manual override on each damper's actuator for the technician.

8 · Decisions (owner) and the engineers' checks

- **Owner (9 Nov 2026):** L5 adopted (V-c + F3; the fall-back F2 if the engineer refuses the beam additions); no HRV; two paths; no tank on the tower; the gallery grille a sealed provision; the cooler an option; render and paint; the two switches and the dust relay; the west vent re-routed; the hood's make-up inlet mandatory.
- **Structural engineer:** the core's walls and pad, the ties to the slabs, B3 / B4 for 4.33 m, R3 widened to 0.40 over y 6.5–8.8, the new beam R-L5 (1.55 m) under the head's south wall, the head frame for wind and **earthquake** (`studies/structure.typ`).
- **MEP engineer:** the fan, the filters, the silencer, D1 / D2 and the dust relay, the louvres and grilles, the hood's make-up inlet, the vent re-route, the chamber's gully and electrics; the breeze estimate confirmed (a CFD or a wind-tunnel check if wanted).
- **Owner's site work:** the on-site weather station (≥ 6 m, 4–6 weeks) confirms the wind faces before the head is detailed.

7 · Maintenance (all from the carved door at ground level; no routine task enters a shaft or climbs a ladder)

Item	What · how	Every	If neglected
the dust trap	lift the grating, sweep / vacuum the basin, hose to the gully	month; week in Mar–Apr	dust blows back into the courtyard
the pre-filter	slide out, wash, dry, back	month (the gauge)	the fine filter clogs early
the fine filter	replace	≈ 6 months (200 Pa)	less fresh air, more fan power
the gully trap	fill / flush (the trap primer)	month	odours from the drain
the outlet louvres, the grilles	brush, vacuum	3 months	dust on the louvres
D1, D2, the dust relay	stroke test, seals; the sensor's lens	year (before spring)	a damper stuck, no dust safety
the fan and the silencer	listen, check the mounts	year	noise
the head's mesh and flaps	**the roof hatch only:** lower the frames on their cords, brush, raise; free the flaps	year; after a dust storm	birds and leaves in the shaft; a quarter short-circuits
the cooler (if bought)	tank and pads cleaned; drained and parked for winter	week in summer; October	scale, smell, a wet chamber

The shaft is smooth-rendered and self-cleaning (dust falls to the trap); nothing requires entering it. The gallery grille's sealed panel is checked with the dampers.

Sources: `studies/malqaf_L5.typ` (Phases 0–6: the climate basis, the geometry, the structure, the facade, the air system, the head, the decision sheet), `studies/roof.typ` (Z2b, the attic, the walkways), `studies/structure.typ`, `studies/plumbing.typ` (the vents), ASHRAE 62.1 / 62.2-2022, ASHRAE Fundamentals 2021 ch. 21 and 24, ISO 16890, EN 1751. Concept only.